



Gowin SDRAM Controller HS IP User Guide

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Revision History

Date	Version	Description
09/07/2020	1.0E	Initial version published.
11/10/2023	1.0.1E	The I_sdrc_addr description of “Table 5-2 User Signals Description” in “5 Port List” optimized.
05/09/2025	1.0.2E	A note added in 8.2 Application .
10/17/2025	1.1E	Descriptions of chapter 8 Application Example updated.

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1 About This Guide

1.1 Purpose

Gowin SDRAM Controller HS IP User Guide mainly includes the features, diagram, principle, signal definition, parameter, GUI, and interface timing. It aims to help you quickly understand the features and usage of Gowin SDRAM Controller HS IP to accelerate product development.

1.2 Related Documents

The latest user guides are available on the GOWINSEMI Website. You can find related documents at www.gowinsemi.com.

- [DS100, GW1N series of FPGA Products Data Sheet](#)
- [DS117, GW1NR series of FPGA Products Data Sheet](#)
- [DS102, GW2A series of FPGA Products Data Sheet](#)
- [DS226, GW2AR series of FPGA Products Data Sheet](#)
- [DS981, GW5AT series of FPGA Products Data Sheet](#)
- [DS1103, GW5A series of FPGA Products Data Sheet](#)
- [DS1239, GW5AST series of FPGA Products Data Sheet](#)
- [DS1105, GW5AS series of FPGA Products Data Sheet](#)
- [DS1108, GW5AR series of FPGA Products Data Sheet](#)
- [DS1118, GW5ART series of FPGA Products Data Sheet](#)
- [SUG100, Gowin Software User Guide](#)

1.3 Terminology and Abbreviations

The terminology and abbreviations used in this manual are as shown in Table 1-1.

Table 1-1 Abbreviations and Terminology

Terminology and Abbreviations	Meaning
FPGA	Field Programmable Gate Array
IP	Intellectual Property
SDRAM	Synchronous Dynamic RAM

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly by the following ways.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

2 Overview

SDRAM is widely used for its high-cost performance. As SDRAM has strict timing requirements and complex logic control, a dedicated controller is required to implement the data read/write, and control functions, etc.

Gowin SDRAM Controller HS IP meets the needs of FPGA users by reducing the complexity of system development and decreasing product development time. The SDRAM is located between the SDRAM and the user logic modules. It offers a user-friendly interface to realize SDRAM data read/write timing control and automatic refresh control.

After calling through the software interface and parameters configuration, Gowin SDRAM Controller HS IP can be applied in Gowin FPGA products after synthesis and P&R through Gowin Software.

Table 2-1 Gowin SDRAM Controller HS IP Overview

Gowin SDRAM HS IP	
Logic Resource	See Table 3-1
Delivered Doc.	
Design Files	Verilog (encryption)
Test and Design Flow	
Synthesis Software	GowinSynthesis
Application Software	Gowin Software

Note!

For the devices supported, you can click [here](#) to get the information.

3 Features and Performance

3.1 Features

- Configurable width of SDRAM data, address, row/column, etc.
- Users can flexibly control over SDRAM refresh
- Users can flexibly control whether read or write pre-charging is enabled
- Supports SDRAM read and write operations in the range of 1~page length
- Users can flexibly control self-refresh and power-down functions
- Supports industry-standard SDRAM devices
- Applies to both on-chip and off-chip SDRAM memory
- Can be synthesized
- The language is Verilog

3.2 Performance

The frequency of Gowin SDRAM Controller HS IP depends on the clock frequency of SDRAM and the maximum frequency supported by the IP on the selected chip.

3.3 Resource Utilization

Table 3-1 Performance Reference

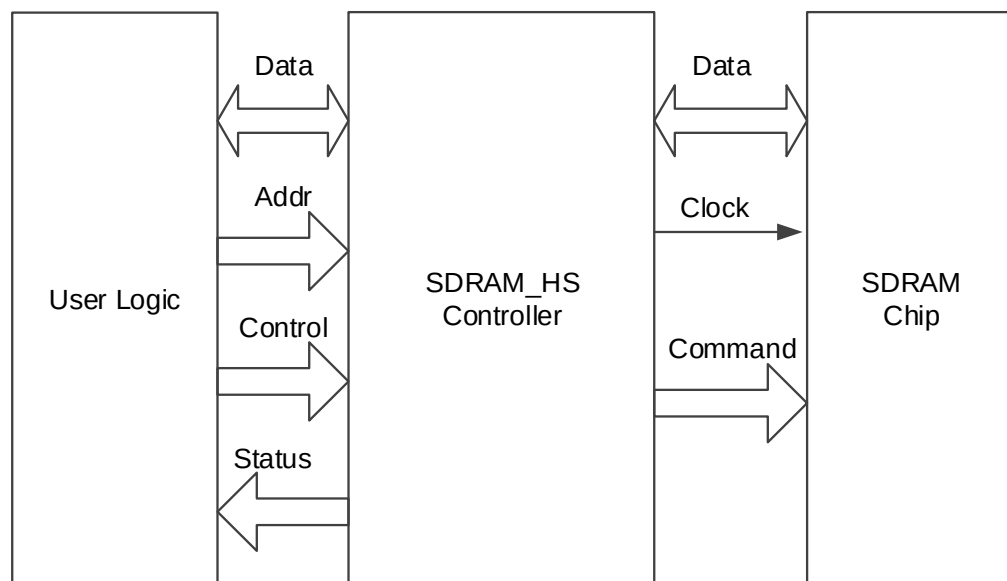
Part Number	Language	LUT4	REG
GW2A-LV18PG256C7/I6	Verilog	≤210	≤170

4 System Block Diagram

4.1 System Block Diagram

Gowin SDRAM Controller HS IP is located between the SDRAM and the user logic modules, as shown in Figure 4-1. The signals that are available on the user logic side mainly consist of data signal, address signal, control signal, status signal, etc. The signals on the SDRAM side mainly include data signal, address signal, and command signal.

Figure 4-1 System Block Diagram



5 Port List

5.1 SDRAM Signals

Table 5-1 Description of SDRAM Signals

No.	Signal Name	I/O	Description	Notes
1	O_sdram_clk	I	Clock	The I/O direction of all the signals takes SDRAM as reference.
2	O_sdram_cke	I	Clock Enable	
3	O_sdram_cs_n	I	Command Control	
4	O_sdram_cas_n	I		
5	O_sdram_ras_n	I		
6	O_sdram_wen_n	I		
7	O_sdram_dqm	I	Data mask control	
8	O_sdram_addr	I	Address	
9	O_sdram_ba	I	Bank	
10	IO_sdram_dq	I/O	Data	

Note!

- When using the SDRAM embedded in the chip, be sure to keep the SDRAM side signal definition and the signal name in Table 5-1 the same.
- When using SDRAM external to the chip, be sure to keep the signal definition on the SDRAM side different from the signal name in Table 5-1.

5.2 User Signals

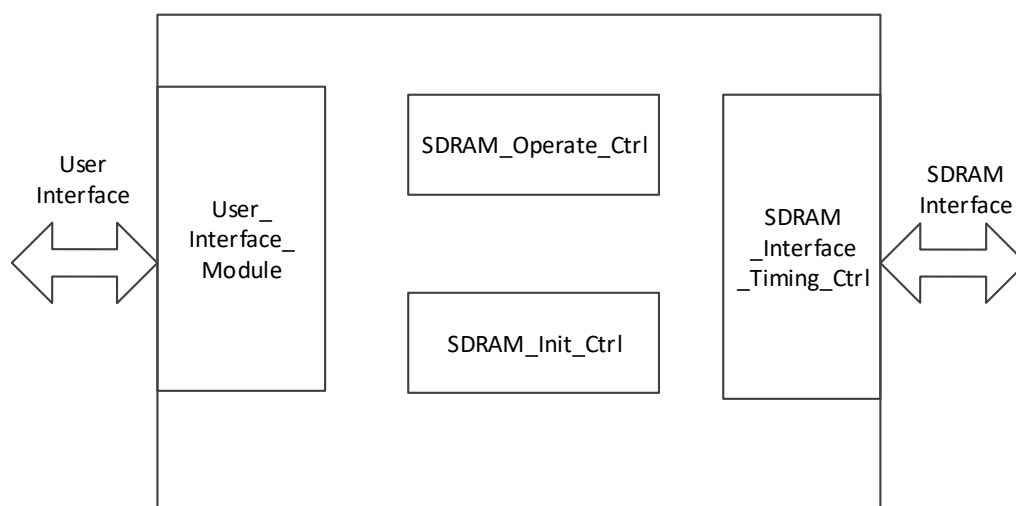
Table 5-2 User Signals Description

No.	Signal Name	I/O	Description	Notes
1	I_sdrc_rst_n	I	Reset signal, active low.	The I/O direction of all the signals takes controller as reference.
2	I_sdrc_clk	I	SDRAM_Controller_HS working clock	
3	I_sdram_clk	I	SDRAM operating clock	
4	I_sdram_selfrefresh	I	Self-refresh Signal request 1: self-refresh enable 0: disable	
5	I_sdram_power_down	I	Power-down request 1: Power-down enable 0: disable	
6	I_sdrc_cmd_en	I	Write enable, Active-high, one clock cycle of pulse width	
7	I_sdrc_cmd	I	Control commands, in descending order, correspond to {O_sdram_ras_n, O_sdram_cas_n, O_sdram_wen_n}.	
8	I_sdrc_precharge_ctrl	I	Precharging control 1: enable pre-charging 0: disable	
9	I_sdrc_addr	I	Address, Bank address, row address, and column address in order from high to low.	
10	I_sdrc_dqm	I	Data mask control	
11	I_sdrc_data_len	I	Read/write data length Take I_sdrc_data_len[7:0] for instance, the port input range is 0~255, and the corresponding valid read/write length range is 1~256, and so on.	
12	I_sdrc_data	I	Write data	
13	O_sdrc_data	O	Read data	
14	O_sdrc_init_done	O	Initialization indication, the controller enters the working state after the initialization is completed. 1: done 0: undone	
15	O_sdrc_cmd_ack	O	The signal flag of control command execution completion held high for 1 clock cycle. 1: done 0: waiting	

6 Structure and Functional Description

6.1 Structure

Figure 6-1 Structure Diagram



The IP design includes User_Interface_Module, SDRAM_Operate_Ctrl, SDRAM_Init_Ctrl, SDRAM_Interface_Timing_Ctrl modules.

- User_Interface_Module: Realizes data communication with user and completes the timing control function of the user interface.
- SDRAM_Operate_Ctrl: Parse according to the command sent from the user side and control the completion of the corresponding action.
- SDRAM_Init_Ctrl: Completes initialization of SDRAM after power-on.
- SDRAM_Interface_Timing_Ctrl: Timing control of the interface with SDRAM chip.

6.2 SDRAM Commands

The common SDRAM commands include no operation, active, read operation, write operation, pre-charge, auto-refresh, configuration mode register, etc. See the table below for detailed operations and definitions.

Table 6-1 Definition of the SDRAM Commands

Commands Name	CS#	RAS#	CAS#	WE#	DQM	ADDR	DQ
Command Inhibit	H	X	X	X	X	X	X
NOP	L	H	H	H	X	X	X
Active	L	L	H	H	X	Bank/row	X
Read	L	H	L	H	L/H	Bank/col	X
Write	L	H	L	L	L/H	Bank/col	Valid
Burst Terminate	L	H	H	L	X	X	Active
Precharge	L	L	H	L	X	Code	X
Auto Refresh or Self Refresh	L	L	L	H	X	X	X
Configuration Mode Register	L	L	L	L	X	Op-code	X
Write enable	X	X	X	X	L	X	Active
Write Inhibit	X	X	X	X	H	X	High-Z

Note!

X: No concern; L: Low level; H: High level

6.3 Initialization

The SDRAM is initialized by the SDRAM controller each time they are powered on. When VDD, VDDQ and clock of the SDRAM memory chip are stabilized at the same time, the controller sends a delay of minimum 100us to pre-charge and then performs two auto-refresh cycles. After the auto-refresh cycle is completed, the operating parameters controlled by the SDRAM mode register are configured, and finally the controller enters the normal operating state after waiting for the tMRD delay.

6.4 Refresh

SDRAM refresh options consist of self-refresh and auto-refresh.

Self-refresh operation: The user can start the self-refresh function by pulling up I_sdram_selfrefresh. To exit self-refresh mode, you can pull down I_sdram_selfrefresh.

Auto-Refresh Operation: The user can perform an auto-refresh action by sending an auto-refresh command to the SDRAM controller. The SDRAM controller indicates the completion of this auto-refresh by activeing high with O_sdrc_cmd_ack. The user autonomously controls the auto-refresh interval.

6.5 Active

Before reading and writing to the SDRAM, it is necessary to first

activate the line to be read and written. The user can complete an activation operation by sending the activation command and corresponding Bank/row information to the SDRAM controller. The SDRAM controller indicates the completion of activation with O_sdrd_cmd_ack active-high.

6.6 Read Operation

The user can complete a read operation by sending the Read command and corresponding Bank/row information to the SDRAM controller. The SDRAM controller indicates the completion of this read operation with O_sdrd_cmd_ack active-high.

At each read operation, the user can tell the SDRAM controller via I_sdrd_precharge_ctrl whether the pre-charge operation needs to be performed after the read operation.

6.7 Write Operation

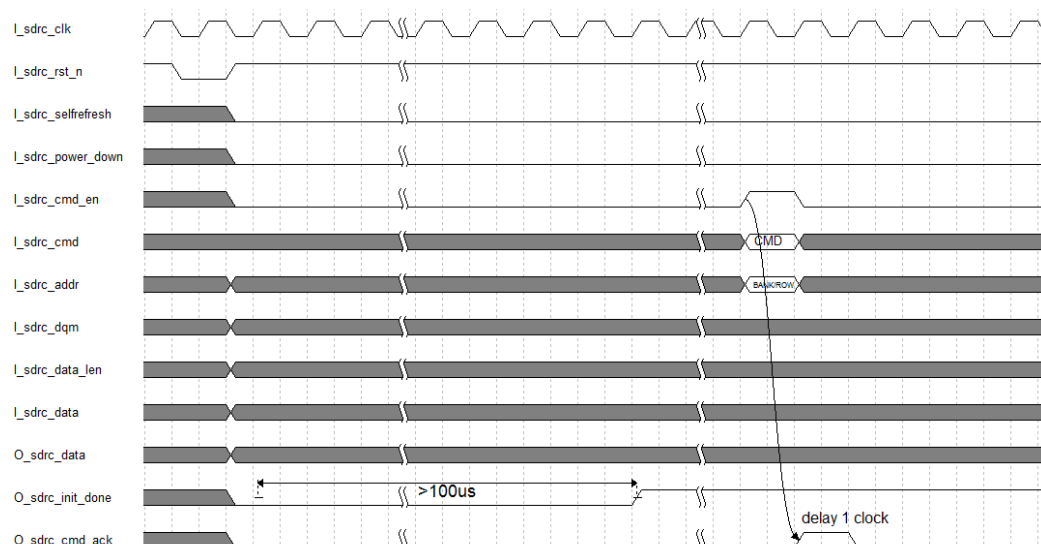
The user can complete a write operation by sending the write command and corresponding Bank/row information to the SDRAM controller. The SDRAM controller indicates the completion of this Write operation with O_sdrd_cmd_ack active-high.

At each Write operation, the user can tell the SDRAM controller via I_sdrd_precharge_ctrl whether the pre-charging operation needs to be performed after the write operation.

6.8 Interface Timing

6.8.1 Activation Operations

Figure 6-2 Activation Operation Timing Diagram

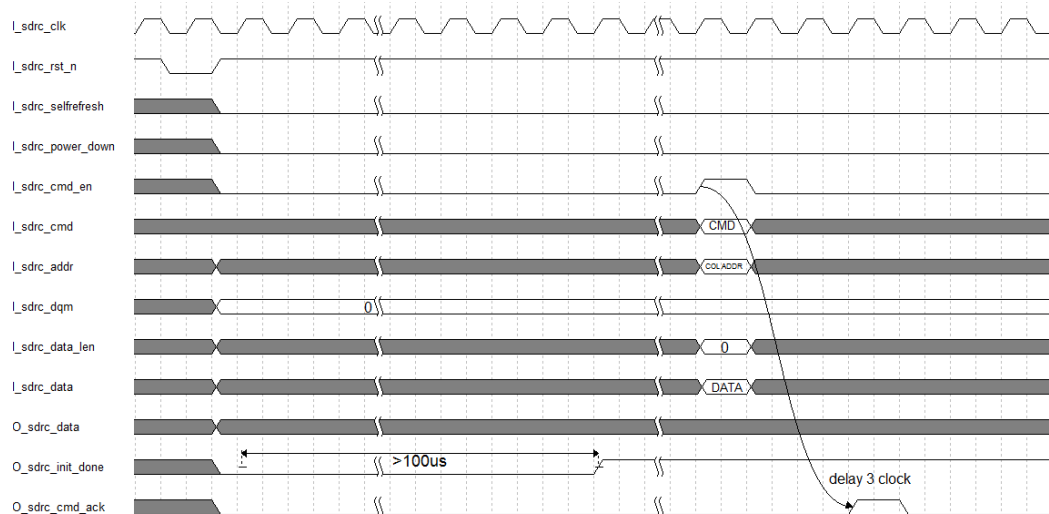


Note!

tRCD=2

6.8.2 Write Operation

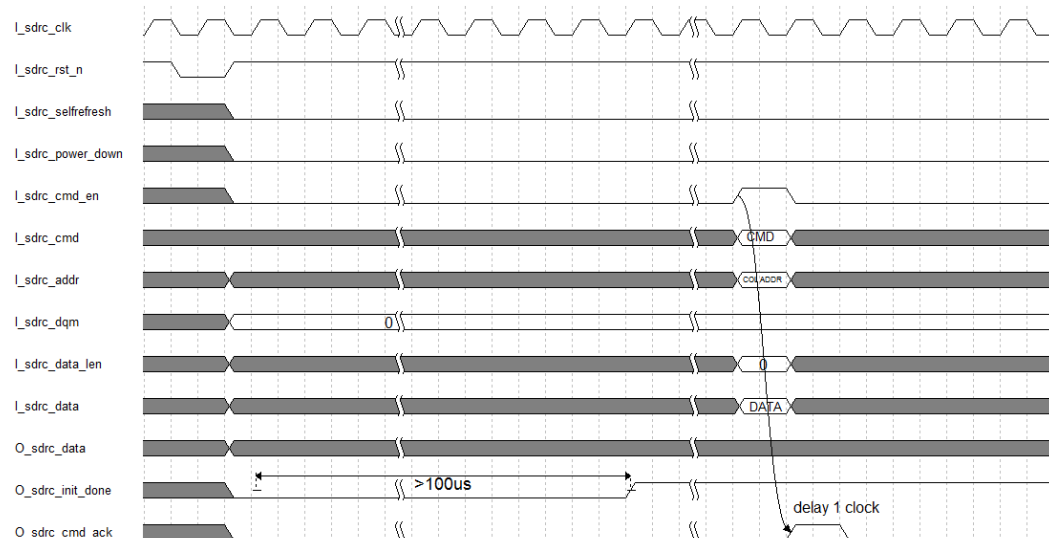
Figure 6-3 Write Operation Timing with Pre-charging



Note!

tWR=2; tRP=2; Readout length = 1.

Figure 6-4 Write Operation Timing without Pre-charging

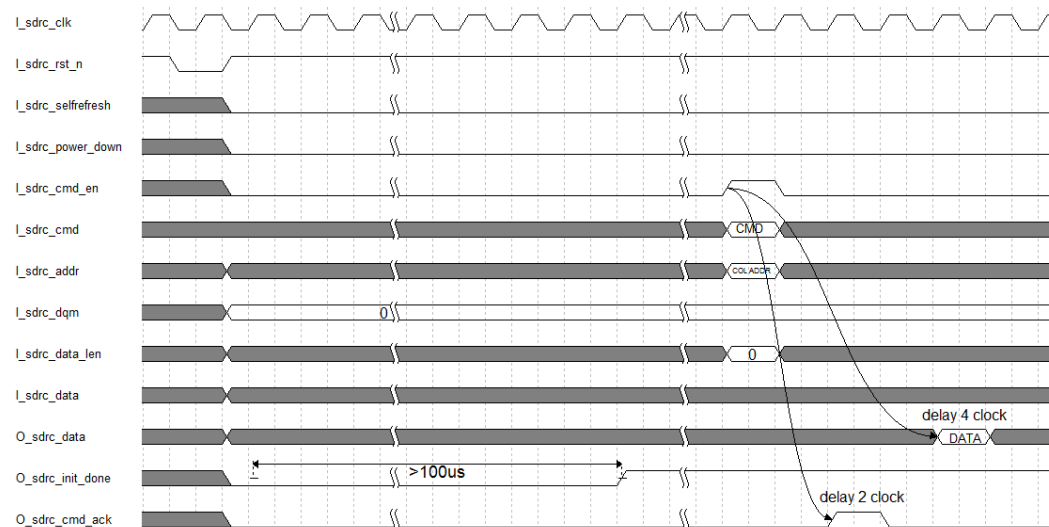


Note!

tWR=2; tRP=2; Readout length = 1.

6.8.3 Read Operation

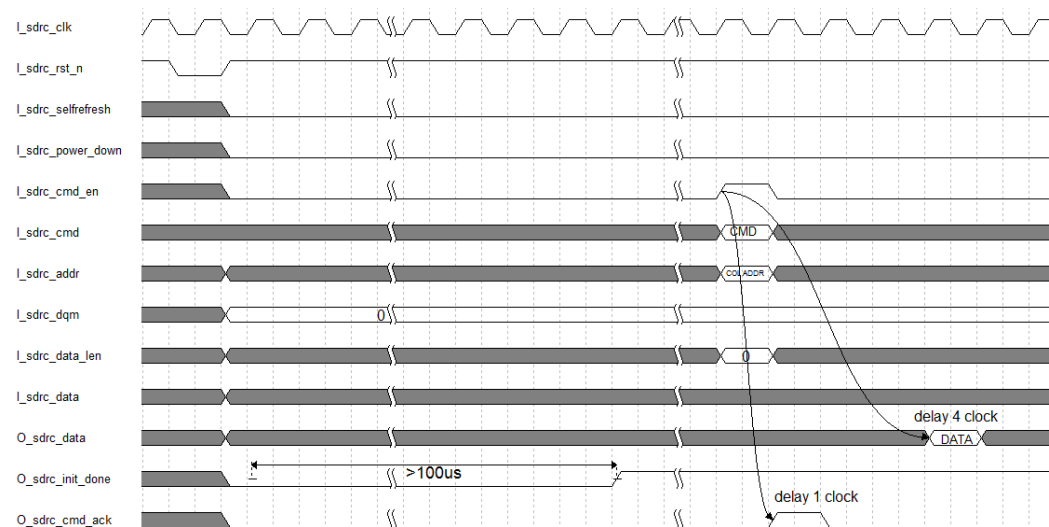
Figure 6-5 Read Operation Timing with Pre-charging



Note!

CL=2; tRP=2; Readout length = 1.

Figure 6-6 Read Operation Timing without Pre-charging

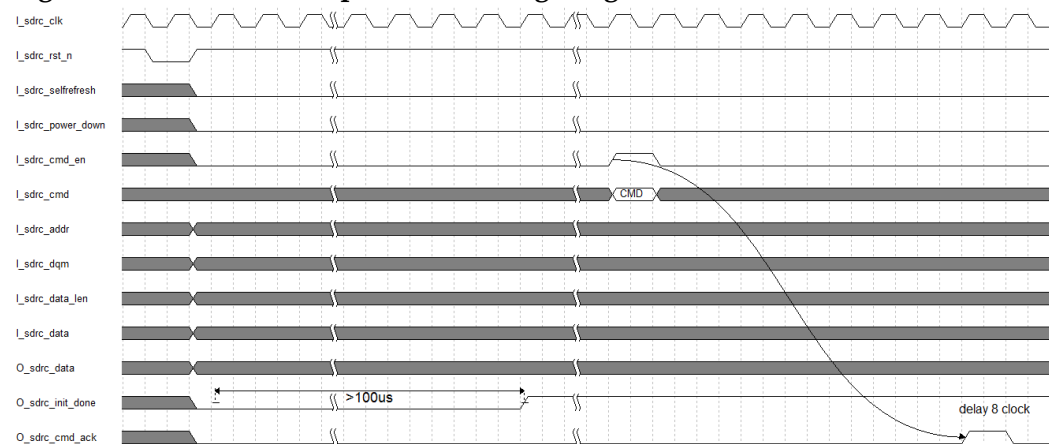


Note!

CL=2; tRP=2; Readout length = 1.

6.8.4 Auto-refresh Operation

Figure 6-7 Auto-refresh Operation Timing Diagram



Note!

$t_{RFC}=9$

7 Parameters Configuration

7.1 Overview

The GUI parameter definitions of Gowin SDRAM Controller HS IP are described in this chapter.

7.2 Parameter

Table 7-1 GUI Parameter Definitions

Data Width	Integral, value range: 8, 16, 32, 64
Bank Width	Integral, Value range: 1, 2
Row Width	Integral, value range: 11~ 14
Column Width	Integral, value range: 8~ 14
CL	Integral, value range: 1~ 3
tRP	Integral, value range: 2~ 4
tRFC	Integral, value range: 9~ 16
tMRD	Integral, value range: 1~ 3
tRCD	Integral, value range: 2~ 4
tWR	Integral, value range: 1~ 3

Note!

The above parameters must be consistent with the parameters required by the selected embedded or external to SDRAM device when setting up the GUI.

8 Application Example

8.1 Overview

This chapter mainly describes how to set parameters through the Gowin Software interface to generate the SDRAM HS Controller module required by the user, and provides guidance on key considerations in practical applications.

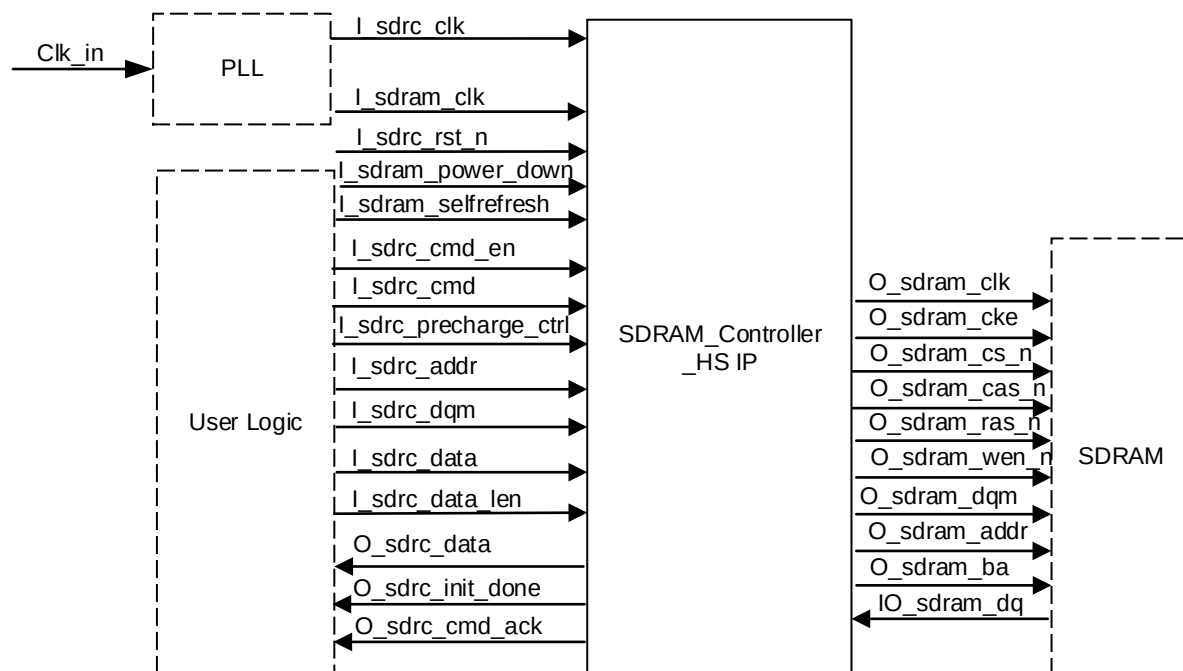
8.2 Application Description

The SDRAM HS Controller application primarily consists of four parts: PLL, user logic, SDRAM HS controller, and SDRAM.

The block diagram of the SDRAM HS application is shown in Figure 8-1.

The PLL is an internal clock resource of the FPGA, which generates the SDRAM operating clock `I_sdram_clk` and the SDRAM HS Controller operating clock `I_sdrc_clk` through clock division or multiplication. When using the SDRAM HS Controller, it is necessary to adjust the phase of `I_sdram_clk` to meet the setup and hold time requirements for read and write signals on the SDRAM side.

All input and output signals of the user logic interface are aligned with the rising edge of `I_sdrc_clk`.

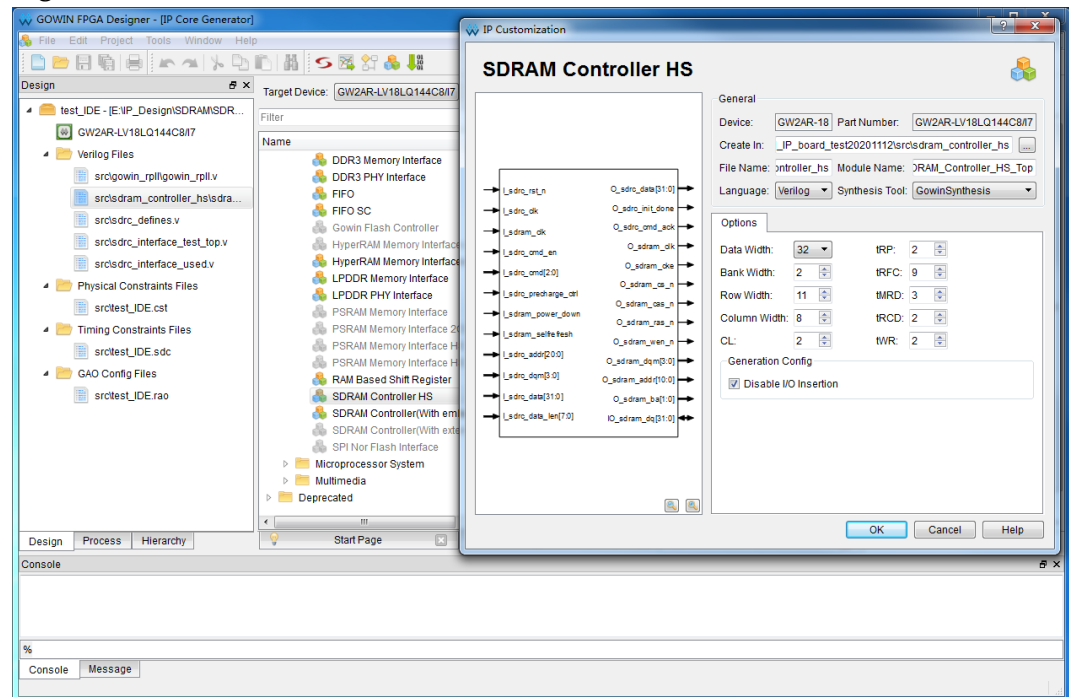
Figure 8-1 Application Diagram**Note !**

- For external SDRAM applications, to improve data transmission performance, the input/output DFFs on the SDRAM side can be constrained to the DFF resources within the IOLOGIC.
- For embedded SDRAM applications, ensure that the SDRAM interface signal names in the top-level design are consistent with the SDRAM-side signal names of the SDRAM HS Controller, and that these signals do not require any constraints in the .cst file.

8.3 Interface Invocation

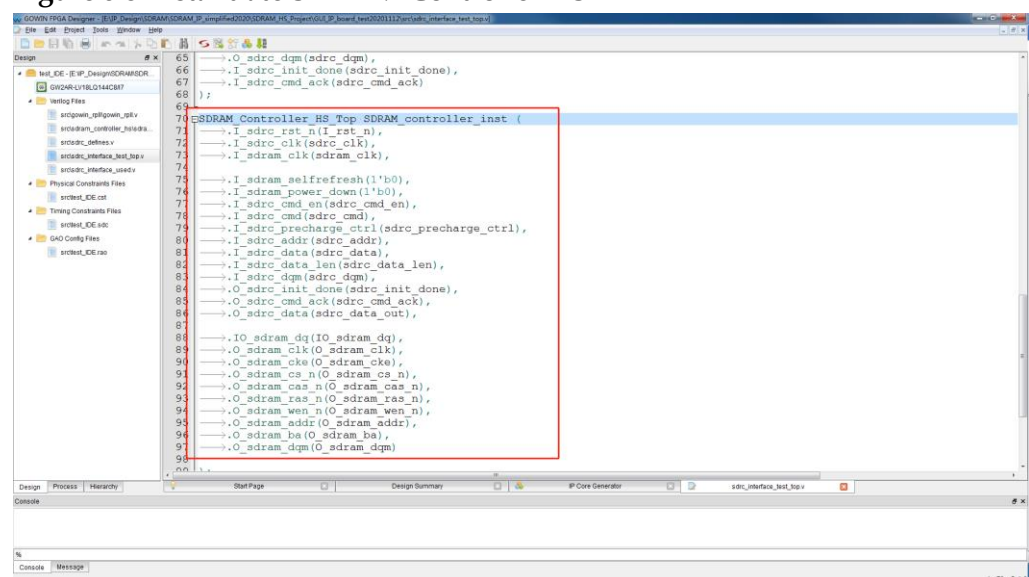
1. After launching Gowin Software, navigate to the "Tools" menu and click "IP Core Generator". Then, select "Memory Control > SDR SDRAM Memory Interface > SDRAM Controller HS" to invoke the "SDRAM Controller HS" IP core. Click "OK" to generate the SDRAM Controller HS Module.

Figure 8-2 Invoke SDRAM Controller HS



2. Instantiate the SDRAM Controller HS in the user program, as shown in the figure below.

Figure 8-3 Instantiate SDRAM Controller HS



3. After synthesis, place-and-route (P&R), and downloading the bitstream file to the FPGA device, functions such as SDRAM data read/write and refresh can be achieved.

